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Assignment - 1

#

1 What is Database Management System (DBMS)? Why is it required?

Answer: A Database Management System (DBMS) is software used to store, manage, retrieve, and organize data efficiently.
It is required because:
- It ensures data security.
- Maintains data consistency.
- Reduce data redundancy.
- Provides backup and recovery.
- Supports multi-user acces.

Ex: Banking system, hospital database

2. Why is DBMS better than the traditional file system?

Answer: DBMS is better because:
- Reduces Redundancy.
- Avoids data inconsistency
- Provides security.
- Supports concurrency
- offers backup & recovery
- Easy data sharing

3. What is Data Abstraction? Explain levels.

Answer: Data Abstraction means hiding internal details & showing only necessary information to users.

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Levels:

1. Physical level: How data is stored.
2. Logical level: Structure of data
3. View level - User-specific data.

4. What is three-schema architecture? Why is it needed?

Answer: it includes:
- internal level.
- Conceptual level
- External level.

Importance:
- Provides data independence
- Simplifies database design.
- Improves security.

5. What is Data Independence? Why is it needed?

Answer: It is the ability to change schema at one level without affecting others.

Types:
- Logical.
- Physical.

→ Needed for flexibility & system maintenance.

6. What is Entity-Relationship (E.R) models?

Answer: The Entity-Relationship model represents real-world objects, attributes and relationships graphically

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7 What are integrity constraints? Why are they important in DBMS?

Answer- Rules that maintain accuracy and consistency.

- Primary key.
- Foreign key
- Unique.
- Not Null.
- Check.

Section - B

Q1 Draw the three - schema architecture and explain each level.

Answer:

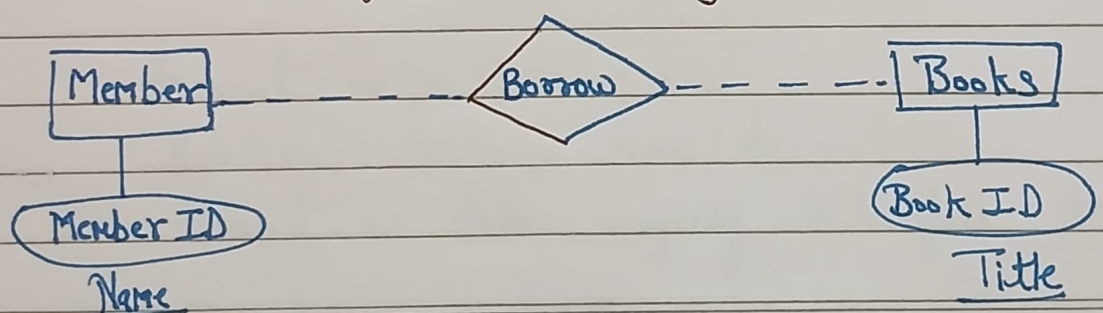
Users
↓
External Level (views)
↓
Conceptual Level (Logical structure)
↓
Internal Level (Physical storage).

Explanation - External : User view.
Conceptual : Overall database
Internal : Storage details.

Benefit : Data independence.

Q2 Draw an ER diagram for a Library Management System.

Answer:



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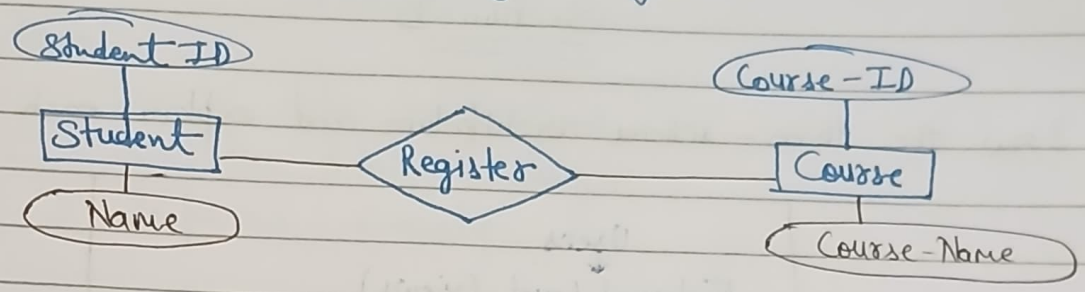
Entities: Member, Book.
Relationship: Borrower.
key: Member ID, Book ID [One member can borrow many books].

Q3 Draw an ER diagram for student-Course Registration System.

Answer Student (student ID, Name)
Course (Course ID, Course-Name).
Relationship: student register for course



Section C

Ques 1 A hospital want to store patient, doctor & appointement data. Identify entities, attributes and relationship.

Answer Entities: Patient, Doctor, Appointement, Patient books Appointement
Doctor attends Appointement.

Primary keys: Patient-ID, Doctor-ID, Appointement-ID.

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An online shopping platform stores customers, product & order data. Suggest keys & constraints.

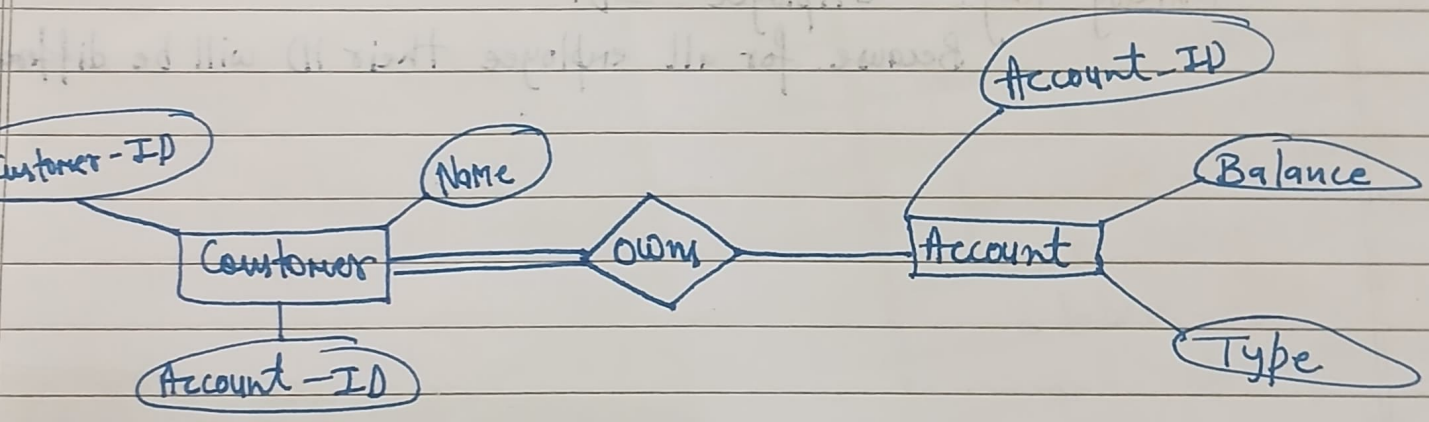
Customer (customer ID - Pk)
Product (Product ID - Pk)
Order (order ID - Pk)
foreign key in order Table connect customer & product

A university stores student records. Explain how abstraction help diff users.

Student see marks (view level)
Admin manager records (Logical level)
DBA controls storage (Physical level).

Section D (ON Insign).

ER Diagram: ATM System.



Account-no is primary key.
Relation: Customer owns Account.
Account no take uniquely Indentifiers Account.

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② Three - Schema Architecture + one Benefit

Answer: Benefit: Provide Data Independence
change is storage do not affect uses.

External Level (user view)
↓
Conceptual Level (logical structure of Database)
↓
Internal Level (Physical storage Level.)

③ Draw a Entity with Attributes.

Entity: Employee.
Employee-ID (PK)
Name
Salary.
Departement.

Primary key: Employee-ID.
Because for all employee their ID will be different

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